PHANTOM

Supplementary Appendix

Nischit Girish Kumar

NEXUS Research Initiative — Independent Research

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Companion to the final technical report and ACM-style manuscript. Historical evidence and confirmed prospective final results are distinguished. This publication supplement includes aggregate evidence and protocol descriptions, not raw source histories or protected evaluator truth.

A. Research chronology

Specification

Objective: Define an event-level research system. Hypothesis: Relational evidence may improve over individual anomaly heuristics.

Method: Architecture, contracts and staged gates. Result: Aspirational specification, not empirical validation.

Gate: Require evidence before stronger claims. Decision and resulting change: Original scope superseded where later falsification narrows it.

1 / data audit

Objective: Establish the source representation. Hypothesis: The release can support reproducible event research.

Method: Disk-backed parquet audit; duplicate, schema and session checks. Result: 704,728,520 events; all source sessions ten records.

Gate: Data use must respect representation limits. Decision and resulting change: Preserve source scale while rejecting unvalidated natural-session assumptions.

2 / behavior

Objective: Characterize bounded listening histories. Hypothesis: Behavioral features can describe variation.

Method: Release-relative user sampling and descriptive analysis. Result: 671,410 sampled rows from 3,252 users.

Gate: Description is not a verified legitimacy label. Decision and resulting change: Use observed histories as a foundation, not labelled negatives.

3 / simulation

Objective: Create controlled coordination tasks. Hypothesis: Synthetic campaigns permit measured recovery.

Method: Six families, three levels, four popularity regimes. Result: Origin and catalog differences were possible.

Gate: Audit generation shortcuts. Decision and resulting change: Introduce neutral and origin probes.

4 / baseline

Objective: Evaluate behavioral and relational candidates. Hypothesis: High performance may represent coordination.

Method: Logistic baselines and graph comparisons. Result: Apparently useful detection coexisted with session and origin problems.

Gate: Benchmark interpretation rejected. Decision and resulting change: Do not promote initial scores into final headline results.

4.5 / remediation

Objective: Remove known session leakage. Hypothesis: Session-free behavior might remove the mismatch.

Method: Actor-weighted session-free experiments and permutation probes. Result: Neutral alerts 12.08% versus observed-background 4.79%.

Gate: NO-GO: residual benchmark shift. Decision and resulting change: Demand stronger source and catalog controls.

4.6 / realism

Objective: Test calendar/catalog origin effects. Hypothesis: More realistic generation could admit relational comparison.

Method: Augmented-origin gates for 72 candidates. Result: All 72 failed admissibility gates.

Gate: No accepted detection evaluation. Decision and resulting change: Move toward within-history controlled intervention.

4.7 / counterfactual

Objective: Pair intervention and sham on observed histories. Hypothesis: Shared source origin might remove shortcuts.

Method: Additive treatment/control construction. Result: Actor marginals still changed.

Gate: NO-GO under required invariance. Decision and resulting change: Preserve marginals rather than merely adding matched rows.

4.8 / structure preservation

Objective: Hold item/time marginals fixed. Hypothesis: Reassignment can isolate cross-actor structure.

Method: Within-actor assignment changes and duplicate audits. Result: 20/180 pairs changed semantic duplicate counts.

Gate: Processing gate failed. Decision and resulting change: Repair collisions before interpreting relational scores.

4.9 / relational calibration

Objective: Use collision-safe controlled pairs. Hypothesis: A strong relational ranking may transfer a threshold.

Method: Fixed schedules and a frozen raw threshold. Result: AUC 1.000 but 22/180 validation controls alerted.

Gate: Standalone burden gate failed. Decision and resulting change: Separate ranking from calibrated threshold behavior.

4.10 / calibration

Objective: Find transferable global calibration. Hypothesis: A global development threshold might satisfy all blocks.

Method: Three declared calibration procedures. Result: All failed per-block ceiling requirements.

Gate: NO-GO despite useful pooled results. Decision and resulting change: Normalize against each eligible world's conditional null.

4.11 / local surprise

Objective: Freeze self-normalized relational detection. Hypothesis: Local baseline and dispersion can improve transfer.

Method: 4095 permutations; robust median/MAD; threshold three. Result: Development separation with no observed control alerts.

Gate: Accepted as frozen candidate; external evidence still needed. Decision and resulting change: Carry identical method to an independent source population.

4.12 / external validation

Objective: Evaluate frozen detector in ListenBrainz. Hypothesis: Controlled relational signal transfers across populations.

Method: 42 groups, three construction seeds and three difficulties. Result: 378/378 treatment alerts; 0/378 controls; AUC 1.000.

Gate: Bounded external-population test passed. Decision and resulting change: Qualify transfer by synthetic intervention and narrow selector.

5 / graph intelligence

Objective: Recover member/target objects. Hypothesis: Temporal graph evidence can reconstruct campaigns.

Method: Pair-level nulls, components and variants. Result: Duplicate sensitivity and unacceptable control burden.

Gate: Graph-null remediation required. Decision and resulting change: Use semantic presence and stronger conditional candidate pooling.

5.1 / graph-null remediation

Objective: Fix representation and pooling. Hypothesis: Fixed catalog pools can aggregate weak relational evidence.

Method: Burst-conditioned null, 60/300 consensus, fragment links. Result: Historical internal member/target F1 0.9167; 0/84 controls.

Gate: Accepted frozen core within support. Decision and resulting change: Retain source identity and test hard boundaries.

6 / hybrid

Objective: Improve actionability with behavior. Hypothesis: Behavioral veto may reduce unwanted alerts.

Method: Hybrid fusion around accepted core. Result: Actionable F1 fell to 0.5357.

Gate: Rejected degradation. Decision and resulting change: Do not replace core with a weaker veto system.

6.1 / non-veto

Objective: Add information without suppressing core. Hypothesis: Non-veto context can improve ranking.

Method: Within-block ranking, stability and cost tests. Result: Zero added gain over 210 comparisons.

Gate: Rejected augmentation; retain core only. Decision and resulting change: Prioritize failure characterization over unsupported complexity.

7 / robustness

Objective: Map perturbation and hard-negative boundaries. Hypothesis: Moderate missingness and negative contexts can be tolerated.

Method: Frozen core on reused development; separate perturbations. Result: Several moderate settings pass; shocks and shifted overlap fail.

Gate: Overall operating region failed. Decision and resulting change: Seek common-cause/overlap explanation and keep failures visible.

7.1 / common cause

Objective: Repair shocks and overlap. Hypothesis: Conditioning or parent association can resolve the failures.

Method: Alternative conditioned nulls; combined gates; causal twins. Result: Partial fixes conflict; identical-input causes remain indistinguishable.

Gate: Combined requirements not met. Decision and resulting change: Recognize missing causal information instead of tuning it away.

7.2 / reframing

Objective: Define a scientifically defensible task. Hypothesis: Structure can aid investigation without intent labels.

Method: Operating, terminology and provenance contracts. Result: Event-only attribution explicitly withheld.

Gate: Govern later work by the narrower claim. Decision and resulting change: Separate support, alternatives, evidence and external provenance.

8 / red team

Objective: Challenge investigative output. Hypothesis: Structural candidates can support stable review.

Method: Synthetic bridge, overlap, ranking and packet attacks. Result: Parent recovery, identifier ranking and packet completeness fail.

Gate: Investigation remediation required. Decision and resulting change: Specify complete evidence and invariant review semantics.

9 / investigation remediation

Objective: Make outputs auditable. Hypothesis: Complete packets and support checks improve interpretability.

Method: 17-section evidence contract and ranking tests. Result: Packets improve; shared-response ranking and support disagreements remain.

Gate: No unqualified supported claim. Decision and resulting change: Separate review tiers from causal conclusions.

9.1 / triage

Objective: Route ambiguous evidence. Hypothesis: Tiered routing can preserve valid review cases.

Method: Queue and hard-response gates. Result: 18/19 gates pass; shared-response handling fails.

Gate: Remediation remains required. Decision and resulting change: Replace unjustified scalar preference with complete equivalence sets.

9.2 / equivalence

Objective: Preserve unresolved alternatives equally. Hypothesis: Representation-relative sets avoid false preferences.

Method: Equal priority, invariant display and complete filtering. Result: 480 candidates; 327/477 original parents reviewable; OOS 40/44.

Gate: Accepted implementation scope; unresolved support disagreements retained. Decision and resulting change: Freeze cautious evidence packets; no fresh population claim.

10 / dashboard

Objective: Expose frozen evidence faithfully. Hypothesis: Read-only interface can preserve the research contract.

Method: Five synthetic runs, 22 hypotheses, academic exports. Result: 3943 engineering checks recorded as passing.

Gate: Engineering freeze, not scientific or analyst validation. Decision and resulting change: Use the frozen interface as an inspectable research demonstration.

11 / Pre-final freeze

The scientific system and PRE-FINAL manuscript were frozen before the final evaluation. The original versions and their provenance snapshots remain byte-identical. Comparative baselines, broader support coverage and analyst studies remained unresolved; a future final result was not permission to retune the method.

12 / Prospective sealed final evaluation and verification lineage

The original final commitment was invalidated before scoring because final412 lacked executable semantics. Phase 12A added one deterministic mapping and resealed before final histories were accessed. A first execution lock was superseded, after toy/non-final equivalence checks and before final event access, to bind subsequently completed execution adapters. The canonical scientific run produced perfect controlled separation. The original verifier completed replay but failed while serializing a NumPy boolean in its JSON receipt. The invalid result and failed attempt remain permanent historical records.

Phase 12B independently checked immutable original/replay artifacts. It reproduced every provisional final metric exactly and confirmed all 756 worlds and 18 applicable scientific gates. It did not rerun scientific producers. Two separate forensic tooling incidents remain documented below.

13 / Final synthesis and bounded release preparation

Final publications incorporate the confirmed result and the complete procedural qualification. Figures use frozen aggregates; arithmetic and byte-identity checks create no new scientific evaluation. A sanitized documentation/synthetic-demo candidate is separate from the private review archive. No publication, deployment, submission or indexing occurs. PHANTOM v1 science is closed.

B. Methodological appendix

B1. Read the two detectors separately

The Phase 4.11 population detector and Phase 5.1 reconstruction core are different methods. The former selects one exact recurring actor set, computes population minute-bin coactivity and uses 4095 null draws. The latter selects singleton-user burst support, constructs a fixed catalog graph and pools component evidence over 511 draws. External ListenBrainz success belongs to the former. Historical internal graph reconstruction belongs to the latter. Their evidence must not be merged into a claim of externally validated end-to-end investigation.

B2. Population detector pseudocode

1. Read only the detector-visible canonical event array. Group exact timestamps by recurring actor set. Require at least eight distinct users occurring once, exactly 32 recurring timestamps in distinct minute bins, span at least 27 days, and exactly one qualifying group. Otherwise return unsupported.

2. Reduce background to unique (user, item, minute) presences for sufficient counts. Keep background, actor identities and selected timestamps fixed. Independently permute each selected actor's sorted item multiset over its slots using 4095 draws and seed 41101.

3. Compute the same ordered cross-actor coactivity numerator and opportunity denominator for observation and every draw. Center by null median and scale by 1.4826 times MAD. Zero scale maps equal/above/below median to 0/positive sentinel/negative sentinel. Alert at robust excess at least three.

4. Return score and support information. This does not return a Gaussian probability, a fraud probability, a campaign organizer or an estimate of natural-world false positives.

B3. Reconstruction pseudocode

1. Restrict the input to START=1651363200 and END=1653782400, with END exclusive. Canonicalize unique (user,item,timestamp) atoms; retain multiplicities as description. Ambiguous actor fingerprints produce abstention.

2. Form bursts by consecutive gaps no greater than 600 seconds. Retain bursts having at least eight distinct users and eight singleton-user atoms. Keep actors present in at least eight qualifying bursts, apply that filter once, then keep groups of at least two selected atoms.

3. Build an untimed selected-catalog graph using shared-item count at least three and Jaccard at least 0.10. Connected components of size at least three are a null-invariant candidate family.

4. Independently permute selected items within each actor with 511 draws and seed 55101. Sum same-item same-bin evidence over shared slots, use the minimum 60/300 count per pair, and average over every

unordered pair within each candidate, including zeros.

5. Normalize observed and null component scores with their common median/MAD. Compare observed scores to the per-draw maximum across fixed components with the add-one denominator 512. Require robust excess at least three and comparison value at most 0.01.

6. Link admitted components with target Jaccard at least 0.25 and overlap at least 0.5 relative to the shorter period. Preserve constituent scores. Targets shared in two member catalogs need not have direct temporal support. Return evidence and withheld causal attribution.

B4. Metrics and uncertainty

For a matched predicted member set P and original parent T, precision is intersection size divided by |P|, recall divides by |T|, and F1 is twice the intersection divided by |P|+|T|. The historical evaluator uses one-to-one Hungarian matching by member F1; unmatched parents receive zero. Mean F1 is the average of per-parent scores, not a recomputation from two unrelated means. Target and fragment metrics have their own denominators. The fragmented-regime recovery claim does not describe every campaign type.

World alert burden counts any structural alert in a control world. It is not a real-world false-positive rate because observed users lack verified legitimacy labels and controls encode a research construction. AUC separates score ranking from the frozen threshold. Primary robustness scores preserve original truth under dropout; surviving-truth analyses answer a different question.

Whole-source resampling respects one layer of clustering but does not establish source independence or mechanism coverage. Perfect observed group outcomes produce degenerate percentile intervals. No population zero-error statement is warranted. The illustrative 42-independent-group zero-event upper bound in the report is explicitly hypothetical, not a replacement confidence interval.

B5. Causal-twin assumptions

The equality-of-output argument requires the same detector-visible input and, for a randomized method, the same randomness law independent of the hidden cause. It does not require the two causal stories to be equally likely. It does not establish posterior odds, or deny that external provenance or well-justified assumptions could identify causes. The six constructed twins demonstrate a limitation of this information setting. Any future provenance extension changes the task and needs new validation.

C. Reproducibility appendix

C1. Historical runtime and source identity

The research relies on Python, NumPy, pandas/DuckDB/parquet workflows, scikit-learn in early behavioral evaluation, SciPy matching and NetworkX in bounded reconstruction, plus browser JavaScript/CSS/HTML for the frozen dashboard. Exact historical versions are recorded in the phase-specific environment receipts; the linked source map includes their identities. They are not replaced with the document-building runtime.

The accepted pooling code SHA-256 is 502267217e01ed904ec1ed7e12088d6b6a062d555819f3fd4cc8c5929a764979. It imports the frozen detector module. A matching file digest does not prove methodological correctness, and running newer dependencies is not automatically a faithful scientific reproduction.

C2. Seed discipline

Population local null: 4095 draws, seed 41101. ListenBrainz sampling: 41201. External construction: 41211, 41212 and 41213. Historical group bootstrap: 2000 replicates, seed 41102. Graph conditional null: 511 draws, seed 55101. Each number belongs to a specific layer; do not substitute them or use them to claim 4095 independent source populations.

C3. Permitted reproduction

This package supports rebuilding figures from frozen aggregate tables, compiling manuscripts, checking citations/claim references, and running a wholly synthetic toy illustration. It does not grant access to licensed raw data or protected evaluator truth. Exact full scientific reruns require their historical environment and separately authorized data. The public package excludes final raw records and protected evaluator truth. Phase 13 inspects frozen aggregate results and hashes preserved inputs, without scientific reruns.

Figure 01, 02, 03, 04, 07 and 08 are vector conceptual diagrams of verified operations. Figure 05 uses frozen external counts. Figure 06 transcribes the Phase 7 moderate-gate table. Figure 09 is a direct public-demo Observatory capture using unchanged historical synthetic evidence and the author-selected NEXUS identity. Figure 10 plots the confirmed September final aggregate counts and all 42 block rates. Figure metadata identifies provenance and alt text; no generative imagery serves as scientific evidence.

C4. Build and integrity

The technical report and conference manuscript have editable LaTeX sources and share a verified BibTeX file. The figure script requires NumPy and Matplotlib; the supplement/brief renderer uses ReportLab and pypdf for inspection. The build receipt records the actual document environment and compiler/class versions. The Phase 13 manifest hashes final payload files, excluding itself and temporary previews. A separate before/after inventory checks preserved Phase 11 and Phase 12 lineage bytes. Historical inputs remain unchanged.

D. Claim and citation navigation

Claims C01-C05 describe the data audit; C06-C13 the controlled external validation; C14-C21 internal graph evidence; C22 causal twins; C23-C27 rejected augmentations and robustness; C28-C30 the investigation layer; C31-C37 the dashboard; C38-C40 earlier benchmark failures. Additional entries transcribe supporting report rows and implementation parameters. Every entry supplies exact wording, source locator, denominator, population, status and scope restrictions.

Citation verification covers 18 primary-source identities and their limited supporting statements, including BAG (AAAI 2026) and Graph Evidential Learning (KDD 2025). RecSys 2026 is used only as an eight-page format reference; that conference has already occurred. SAGE is in WSDM Companion 2026, not the main WSDM research track. The temporal/multiplex work is labelled a preprint. No PHANTOM DOI, peer-review status, institutional sponsor or publication venue is invented.

E. Public-release boundary

The internal evidence directory contains local source pointers and source excerpts. It is not the public repository. Release only the explicit allowlisted subset after rights, privacy and author sign-off. Do not upload the workspace wholesale. The offline portfolio remains noindex, has no live canonical domain and sends no telemetry. Future deployment requires explicit author authorization and is not performed in Phase 13.

F. Prospective final: definitions, custody and interpretation

F1. Population and treatment/control denominator

The committed source is ListenBrainz snapshot 2723 (October 15, 2026). The final observation window is [2026-09-01 00:00:00, 2026-09-29 00:00:00) UTC. Selection yields 4,032 listeners in 42 groups of 96 with 3,714,102 original observed events. Sampling seed is 41290; construction seeds are 41291, 41292 and 41293. Three levels per seed give 9 pairs per group, 378 pairs and 756 worlds. The May external test instead uses 4,004,802 original events; these counts must not be interchanged.

The null remains 4,095 draws with seed 41101; alert threshold remains robust excess at least 3.0. No post-final detector or null generation was performed in Phase 12B or Phase 13. “Single scientific execution” means one canonical prospective evaluation. The original verification process deliberately reconstructed worlds and replayed nulls as its committed determinism check; those computations are not concealed or counted as independent scientific trials.

F2. AUC and intervals

Pooled AUC is the fraction of all treatment/control score comparisons with treatment greater than control, plus half the tied fraction. Mean block AUC averages the same statistic over 42 groups with equal group weights. Each group has 9 worlds per arm, so its AUC uses 81 comparisons; the pooled AUC uses 142,884 comparisons. These are arithmetic comparison counts, not independent observations.

The fixed threshold yields 378/378 treatment alerts and 0/378 control alerts. Both AUCs are 1.000. Worst-block control burden is 0.0%. The committed 2,000 whole-block resamples, seed 41102, give the conditional AUC and detection intervals [1.000, 1.000], control-mean interval [0.000, 0.000], and equal-block upper 95% control bound 0.0%. Every observed block is perfect, so these degenerate empirical intervals are expected. They do not bound unobserved mechanisms, establish independence across groups or imply a zero natural-population alert probability. No new interval is substituted for the committed procedure.

F3. Ordered procedural record

1. The original v1 commitment fixed the final protocol, but final412 was not executable from the inherited constructor. The original disposition is permanently invalid before scoring.
2. Phase 12A defined explicit ordered hash/RNG inputs for enrollment, catalog and timestamps. The v2 seal was issued at 2026-10-21 05:06:39 UTC, before final history access. Other domains, scientific thresholds and null draws stayed unchanged.
3. The first execution lock (05:13:04 UTC) preceded completed adapters. Header indexing began, but no final event columns were accessed. The superseding lock (05:44:01 UTC) followed toy/non-final equivalence checks and bound the completed execution-file closure. A later population manifest binds back to that lock; the lock was not rewritten to include outcomes.
4. The selected population manifest was recorded at 06:19:27 UTC. The top-level raw-output freeze was recorded at 06:57:47 UTC. The completed independent replay freeze was recorded at 07:31:47 UTC. All times are local machine records, not third-party attestations.
5. The original verifier reached receipt serialization and failed at 07:31:50 UTC. Its partial receipt, traceback and invalid-result record survive. Nine true checks were serialized; later unsaved values are not asserted to have been successfully recorded.
6. Post-final forensic verification used immutable original/replay artifacts and independent metric arithmetic. It reproduced every provisional metric exactly, including 42 block records, 126 block/level cells and three level summaries, and confirmed the original 18 scientific gates. The additional reproduction requirement is satisfied by post-final verification; the original status remains NOT EVALUATED in its historical ledger.

F4. Exact receipt defect and forensic tooling incidents

The failing field was checks.denominators_worlds_pairs_groups. Its conjunction ended in sum(y) == 378 on a NumPy integer array, yielding numpy.bool / numpy.bool_ rather than native bool under NumPy 2.0.2. Standard JSON serialization raised TypeError: Object of type bool is not JSON serializable. The locked verifier was not patched. A separate writer converts only NumPy booleans to native booleans and retains errors for other unsupported types.

Forensic attempt 1 encountered a capability-guard denial during NumPy lazy initialization. A new tool version initialized that library before the guard; seeded metric arithmetic was unchanged. Attempt 2 reproduced all metrics but an added check incorrectly expected evaluator null caches inside the constructor-only reconstruction directory. The retained discrepancy receipt and source definitions identified the actual layout. Receipt completion checked 378 truth-file hashes and 1,512 exact cache comparisons across original/replay conditions, reusing completed metric computations. No discrepancy was hidden; no scientific equality rule was relaxed.

All 756 original/replay world identifiers and expected artifacts were confirmed. The separate verifier generated no new assignments, predictions or null draws. Protected truth arrays were hashed, not deserialized. Independent means separate calculation/process, not an external institution. Hashes establish bytes, not the absence of all possible unrecorded activity.

F5. Key immutable identifiers

Governing execution lock: fda63407156d072ed2cfddfcdcb46940134a021d953713a883a614f9399dc5c5.

Original detector-output freeze: 4244dfccc3905ef1a71f5850d90eee0c234a72e92a1cdfe3c7a01af545ff8703.

Completed replay freeze: 9bb1f491937798627c35ffc85a38375f66e4350e4b17552bf4bcf886994e5f70.

Original invalid-result record: 585f91baf7ecfa6d2ec4a8b916b8a95e4933beb1f74a67e957304043ac8418d2.

Corrected forensic receipt: 55de97d8c477b4f81d4744f037d297e8333d77b4347c7c2943391cd15f06611d.

The publication source inventory maps these identities to retained local files. Only sanitized hash identifiers and aggregate descriptions belong in the public package; private paths, raw histories, exclusions identifying real users and protected truth do not.

F6. What remains unestablished

Final labels describe a controlled intervention. Real users are not classified as fraudulent. The final does not establish real fraud accuracy, production false-positive rates, manipulative intent, causal attribution, bot identity, unrestricted coverage, analyst effectiveness, enforcement suitability or production readiness. Causal twins remain counterexamples because successful structural separation cannot add missing causal provenance.

G. Version and reproduction boundary

PHANTOM v1 is the scientific system frozen before the final and evaluated in Phase 12. FINAL v1 is this publication artifact. The dashboard is the unchanged Phase 10 evidence system. A scientific method change requires PHANTOM v2 and fresh, separately authorized evaluation. The September final population is consumed forever for unseen-evidence claims.

Publicly reproducible: equations, figures from aggregate tables, manuscripts and the synthetic teaching pipeline. Authorized-access reproduction: historical population selection, full experiments and raw evaluation, requiring source permissions and exact historical dependencies. Not released: protected final truth, raw

listening histories and private orchestration logs. The public teaching script is not the frozen detector. No general open-source license is selected on the author's behalf.

H. Construction, notation and implementation audit

H1. Controlled-pair construction pseudocode

1. Select a source group with the committed eligibility and exclusion rules. Retain the original event window as the common background. The final selection seed is 41290; this algorithm description does not execute selection.

2. For each committed construction seed and level, use the sealed final412 domain mapping to determine enrollment, catalog and timestamp assignments. Construct the declared recurring schedule with 32 slots and eight items repeated four times. Levels specify 32, 24 or 16 synchronized slots.

3. Produce aligned treatment assignments and the collision-safe matched-control reassignment. Preserve actor identity, row counts, per-actor item and timestamp multisets, and the declared duplicate invariants. Reject a failed invariant; a favorable detector score cannot repair it.

4. Keep arm labels and protected parent information outside the detector-visible array. Score the frozen detector on canonical visible inputs only. The labels describe the construction, not real user intent. Matched arms and seed/level replicas retain shared-source dependence.

This is descriptive pseudocode for the frozen construction, not replacement code. Exact ordering, hash domains and random-number calls are committed in the preserved executable mapping.

H2. Notation checklist

E is the visible event collection; X is the complete permitted input in the causal argument. u and v index actors, i items, t timestamps, b minute bins, and r randomization draws. n_ib counts unique actor presences for one item/bin; N_b counts actor-item presences in the bin; d_ub is the number of distinct items for one actor/bin. The denominator removes same-actor ordered opportunities.

C is the population coactivity statistic. m is its local null median; the scale is 1.4826 times the median absolute deviation. Z is robust excess, not a normal z-test probability. In reconstruction, G is a fixed candidate component, w_uv^(h) counts shared-slot same-item coincidences at width h, and T(G) averages the 60/300 minimum across all unordered actor pairs, including zeros. M^(r) is the maximum normalized candidate score for draw r.

Every threshold and exception in the main methods is descriptive of frozen code. The zero-scale ordering sentinels are not measured effect sizes. The maximum-family comparison is not asserted to prove organic-listening familywise error control. A merged object has constituent evidence, not a newly calibrated merged p-value.

H3. Computational scope

The source audit uses disk-backed scans; its 704.7 million-event scope is not a detector throughput result. The population detector scores fixed-background sufficient counts without copying every complete null world. Graph reconstruction stores draw-by-actor-by-slot assignments plus draw-by-component statistics. Catalog comparisons can still be quadratic in selected actors. No production latency, memory budget or unrestricted stream coverage was measured.

H4. Failure-to-inference map

Origin/neutral/catalog/session failures: generated source or representation features could stand in for coordination; reject the benchmark before accepting detection scores. Duplicate mismatch: preprocessing violated the intended controlled comparison; repair collisions before interpreting performance. Raw/global threshold failure: ranking and alert burden are different claims; local normalization is evaluated separately. Bad graph null: catalog connectivity and incorrect reference conditioning can alert on controls; retain fixed candidates and explicit temporal nulls. Hybrid/non-veto failures: more features need measured benefit and must not erase useful evidence. Shock/shifted-overlap failures: natural common stimuli and overlapping structure defeat a broad operating claim. Causal/ranking ambiguity: absent information cannot be repaired by a stronger scalar score; preserve alternatives and equal-priority sets.

H5. Full final gate definitions

The following criteria are transcribed from the preserved gate ledger, not chosen after results. Gates 1-18 are scientific requirements: their original and post-final ledger outcomes are PASS. Gate 19 is the separate reproduction requirement: its original outcome remains NOT EVALUATED and its post-final outcome is PASS. This does not change the original verifier status, FAILED DURING RECEIPT SERIALIZATION.

1. INTEGRITY: v2 seal/code/config/environment match; original provenance and Phase 11 unchanged.

2. SOURCE: Committed snapshot and final window; >=10 events, >=2 tracks; 42 groups x 96 listeners; exact clones kept together; no incompatible substitution.

3. EXCLUSION: Exclude all 16783 committed May 2022 users and known development clone conflicts before scoring.

4. A: Exact paired behavior; both frozen models mean oriented AUC<=0.55 and worst<=0.60; KS<=0.10 and |SMD|<=0.20.

5. B: Exact original rows/schema/marginals/duplicates/layout; formatting positive-control AUC>=0.95.

6. C0: Each level mean injected excess>=0.10; positive unit fraction>=0.90; every block/seed mean positive.

7. D: Strict ladder fraction>=0.80 and level mean excess strictly decreasing.

8. E: Disjoint source groups; all paired units/levels kept together; exact ladder marginals.

9. each_block_controls: Each block control alert rate<=0.07.

10. block_mean_upper95: Equal-block bootstrap mean upper95<=0.07; 2000 draws, seed41102.

11. each_block_level_detection: Every block/level treatment detection>=0.80.

12. mean_auc: Mean block AUC>=0.70.

13. each_block_level_auc: Every block/level AUC>=0.60.

14. behavior: Frozen behavior gates pass in both duplicate views.

15. block_dependence_reduced: Block eta-squared reduced>=25% from fixed Phase4.10 reference.

16. composition_dependence_reduced: Maximum absolute Pearson reduced>=25% from fixed Phase4.10 reference.

17. science_and_views: All frozen mechanics and full/deduplicated views agree.

18. detector_ladder: Strict detector-feature ladder fraction>=0.80 and decreasing level means.

19. REPRODUCTION: Every world hash matches; independent fresh-process null replay and independent metric verification match.

The scientific verification basis is frozen evidence and unchanged predeclared arithmetic. Reproduction relies on completed immutable reconstruction/replay plus post-final comparisons. The supporting records are PHASE12_CORRECTED_VERIFICATION_RECEIPT.json and evidence/scientific_hash_comparison.json. Their identities appear in the source inventory. The exact machine-readable 19-row ledger, including original and post-final outcomes, is included as tables/final_gate_definitions.csv. Neither the original failed receipt nor its historical outcome is overwritten.

Publication status and authorship

Independent technical report; not peer reviewed or accepted at a conference. Publication and submission remain subject to final author approval.

AI-assisted tools supported selected engineering tasks, with minor assistance in documentation and manuscript preparation. Nischit Girish Kumar directed the research, system architecture, experimental design, validation, and final technical decisions and assumes responsibility for the work.

Source-based acquisition, schema, preprocessing and eligibility instructions are provided in the repository document docs/SOURCE_REPRODUCTION.md. Researchers must obtain datasets from their official sources; this package contains no raw datasets. Exact experimental reproduction also requires separately authorized scientific code and protected evaluator resources.